

Cloud computing

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Assignment

Section B

Getting Started

```
hafsatariq071@ubuntu:~$ cd CC
hafsatariq071@ubuntu:~/CC$ mkdir -p Assignments/Assignment01/screenshots
hafsatariq071@ubuntu:~/CC$ ls
Assignments  Labs
hafsatariq071@ubuntu:~/CC$ ls -la
total 20
drwxrwxr-x 5 hafsatariq071 hafsatariq071 4096 Oct  4 12:29 .
drwxr-x--- 7 hafsatariq071 hafsatariq071 4096 Oct  4 12:27 ..
drwxrwxr-x 7 hafsatariq071 hafsatariq071 4096 Oct  4 12:27 .git
drwxrwxr-x 3 hafsatariq071 hafsatariq071 4096 Oct  4 12:29 Assignments
drwxrwxr-x 3 hafsatariq071 hafsatariq071 4096 Oct  4 12:27 Labs
hafsatariq071@ubuntu:~/CC$ _
```

Task 1: Install Gitea and Push a Repository from the Ubuntu Server

Step 1.1: Install and Run Gitea on the Ubuntu Server

2.

```
hafsatariq071@ubuntu:~$ sudo apt update -y
Hit:1 http://pk.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:3 http://pk.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://pk.archive.ubuntu.com/ubuntu resolute-backports InRelease

** (appstreamcli:2038): WARNING **: 12:48:57.942: No origin found for file pk.archive.ubuntu.com_ubuntu_dists_resolute-updates_includes.yaml.gz
** (appstreamcli:2038): WARNING **: 12:48:57.943: No origin found for file pk.archive.ubuntu.com_ubuntu_dists_resolute-updates_main_source_packages.yaml.gz
** (appstreamcli:2038): WARNING **: 12:48:57.943: No origin found for file security.ubuntu.com_ubuntu_dists_resolute-updates_main_source_packages.yaml.gz
** (appstreamcli:2038): WARNING **: 12:48:57.943: No origin found for file security.ubuntu.com_ubuntu_dists_resolute-updates_main_source_packages.yaml.gz
All packages are up to date.
hafsatariq071@ubuntu:~$
```

3.

```
no VM guests are running outdated hypervisor (qemu) binaries on this host.
hafsatriq071@ubuntu:~$ sudo mkdir -p /usr/local/lib/docker/cli-plugins
hafsatriq071@ubuntu:~$ sudo curl -SL https://github.com/docker/compose/releases/latest/download/docker-compose-linux-x86_64 -o /usr/local/lib/docker/cli-plugins/docker-compose
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left     Speed
  0     0    0     0    0     0      0      0      0      0
100 31.14M 100 31.14M    0     0  3.49M      0  00:00  00:00    3.49M
hafsatriq071@ubuntu:~$ sudo chmod +x /usr/local/lib/docker/cli-plugins/docker-compose
hafsatriq071@ubuntu:~$
```

4.

```
hafsatriq071@ubuntu:~$ sudo systemctl start docker
hafsatriq071@ubuntu:~$ sudo usermod -aG docker $USER
hafsatriq071@ubuntu:~$ exit
```

5.

```
hafsatriq071@ubuntu:~$ git --version
git version 2.53.0
hafsatriq071@ubuntu:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4
hafsatriq071@ubuntu:~$ docker compose version
Docker Compose version v5.6.0
hafsatriq071@ubuntu:~$
```

6.

```
fatal: Authentication failed for 'https://github.com:hafsatriq071/gitea.git/'
hafsatriq071@ubuntu:~$ cd ~
hafsatriq071@ubuntu:~$ git clone https://github.com/WaqasSaleem97/Gitea.git
Cloning into 'Gitea'...
remote: Enumerating objects: 16, done.
remote: Counting objects: 100% (16/16), done.
remote: Compressing objects: 100% (9/9), done.
remote: Total 16 (delta 4), reused 7 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (16/16), 5.37 KiB | 687.00 KiB/s, done.
Resolving deltas: 100% (4/4), done.
hafsatriq071@ubuntu:~$ cd Gitea
hafsatriq071@ubuntu:~/Gitea$
```

7.

```
hafsatriq071@ubuntu:~/Gitea$ docker compose up -d
[+] up 25/25
  ✔ Image postgres:17-alpine Pulled
  ✔ Image gitea/gitea:latest Pulled
  ✔ Network webnet Created
  ✔ Volume gitea_postgres Created
  ✔ Volume gitea Created
  ✔ Container gitea Started
  ✔ Container gitea_db Started
hafsatriq071@ubuntu:~/Gitea$
```

8.

```
hafsatriq071@ubuntu:~/Gitea$ docker compose ps
NAME                IMAGE                COMMAND                SERVICE    CREATED        STATUS        PORTS
gitea                gitea/gitea:latest  "/usr/bin/entrypoint..."  gitea      2 minutes ago  Up 2 minutes  22/tcp, 0.0.0.0:3000
gitea_db             postgres:17-alpine  "docker-entrypoint.s..."  db         2 minutes ago  Up 2 minutes  5432/tcp
hafsatriq071@ubuntu:~/Gitea$
```

9.

```
hafsatriq071@ubuntu:~/Gitea$ curl -s -o /dev/null -w "Gitea HTTP status: %{http_code}\n" http://127.0.0.1:3000/
Gitea HTTP status: 200
hafsatriq071@ubuntu:~/Gitea$
```

10.

```
hafsatriq071@ubuntu:~/Gitea$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:9a:e1:8d brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c299ae18d
    inet 192.168.89.131/24 metric 100 brd 192.168.89.255 scope global dynamic ens33
        valid_lft 944sec preferred_lft 944sec
    inet6 fe80::20c:29ff:fe9a:e18d/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 5a:f4:2f:21:97:47 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
4: br-6c1e41fe7de2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether be:db:f3:f6:cf:52 brd ff:ff:ff:ff:ff:ff
    inet 172.18.0.1/16 brd 172.18.255.255 scope global br-6c1e41fe7de2
        valid_lft forever preferred_lft forever
    inet6 fe80::bcd6:f3ff:fe6:cf52/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
5: vethc839d2101f2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-6c1e41fe7de2 state UP group default
    link/ether 56:d2:b4:c4:ff:03 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet6 fe80::54d2:b4ff:fec4:ff03/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
6: veth8db2db201f2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-6c1e41fe7de2 state UP group default
    link/ether e2:61:52:a4:e8:38 brd ff:ff:ff:ff:ff:ff link-netnsid 1
    inet6 fe80::9485:cbff:fe38:780c/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
hafsatriq071@ubuntu:~/Gitea$
```

12.

If you run Gitea inside Docker, please read the [documentation](#) before changing any settings.

Database Settings

Gitea requires MySQL, PostgreSQL, MSSQL, SQLite3 or TiDB (MySQL protocol).

Database Type *

Host *

Username *

Password *

Database Name *

SSL *

Schema

Leave blank for database default ("public").

13.

▼ Administrator Account Settings

Creating an administrator account is optional. The first registered user will automatically become an administrator.

Administrator Username

Email Address

Password

Confirm Password

These configuration options will be written into: `/data/gitea/conf/app.ini`

To fine tune more details: edit the configuration file after installation. Reference: [app.example.ini](#).

task1 gitea running.png

Step 1.2: Create and Push the Initial Repository

1.

Owner *



hafsatriq071

Some organizations may not show up in the dropdown due to a maximum repository count limit.

Repository Name *

Assignment01

Good repository names use short, memorable and unique keywords. A repository named ".profile" or ".profile-private" could be used to add a README.md for the user/organization profile.

Visibility

☐ Make repository private

Only the owner or the organization members if they have rights, will be able to see it.

Description

Enter short description (optional)

2.

User Settings

Profile

Account

Appearance

Security

Blocked users

Applications

SSH / GPG Keys

Audit Log

Actions >

Packages

Webhooks

Organizations

Repositories

Manage Access Tokens

These tokens grant access to your account using the Gitea API.

Generate New Token

Token Name

ubuntu-server-token

Repository and Organization Access

☐ Public only
 ☒ All (public, private, and limited)

Selected token permissions limit authorization only to the corresponding API routes. Read the [documentation](#) for more information.

activitypub	☒ No Access	☐ Read	☐ Read and Write
admin	☒ No Access	☐ Read	☐ Read and Write
issue	☒ No Access	☐ Read	☐ Read and Write
misc	☒ No Access	☐ Read	☐ Read and Write
notification	☒ No Access	☐ Read	☐ Read and Write
organization	☒ No Access	☐ Read	☐ Read and Write
package	☒ No Access	☐ Read	☐ Read and Write
repository	☐ No Access	☐ Read	☒ Read and Write
user	☒ No Access	☐ Read	☐ Read and Write

Generate Token

3.

```
hafsatarig071@ubuntu:~/Gitea$ cd ~
hafsatarig071@ubuntu:~$ mkdir Assignment01
hafsatarig071@ubuntu:~$ cd Assignment01
hafsatarig071@ubuntu:~/Assignment01$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to 'main' in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:     git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:     git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/hafsatarig071/Assignment01/.git/
hafsatarig071@ubuntu:~/Assignment01$ git branch -M main
hafsatarig071@ubuntu:~/Assignment01$
```

4.

```
hafsatarig071@ubuntu:~/Assignment01$ git branch -M main
hafsatarig071@ubuntu:~/Assignment01$ echo -e "Name : Hafsa Tariq\nReg No : <fa24b1-se-071>" > README.md
hafsatarig071@ubuntu:~/Assignment01$ git commit -m "add student information"
On branch main

Initial commit

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  README.md

nothing added to commit but untracked files present (use "git add" to track)
hafsatarig071@ubuntu:~/Assignment01$
```

5.

```
hafsatarig071@ubuntu:~/Assignment01$ git add README.md
hafsatarig071@ubuntu:~/Assignment01$ git commit -m "add student information"
[main (root-commit) b47fbbd] add student information
 1 file changed, 2 insertions(+)
 create mode 100644 README.md
hafsatarig071@ubuntu:~/Assignment01$ git push -u gitea main
Username for 'http://127.0.0.1:3000': hafsatarig071
Password for 'http://hafsatarig071@127.0.0.1:3000':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 262 bytes | 20.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://127.0.0.1:3000/hafsatarig071/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'gitea/main'.
hafsatarig071@ubuntu:~/Assignment01$
```

7

```
hafsatarig071@ubuntu:~/Assignment01$ git remote add gitea http://127.0.0.1:3000/hafsatarig071/Assignment01.git
```

8

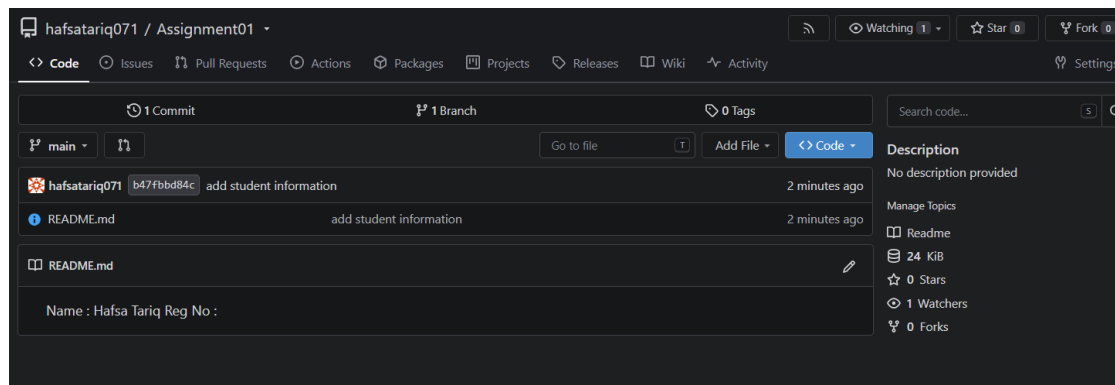
```
hafsatar iq071@ubuntu:~/Assignment01$ git remote -v
gitea    http://127.0.0.1:3000/hafsatar iq071/Assignment01.git (fetch)
gitea    http://127.0.0.1:3000/hafsatar iq071/Assignment01.git (push)
```

9

```
hafsatar iq071@ubuntu:~/Assignment01$ git push -u gitea main
error: src refspec main does not match any
error: failed to push some refs to 'http://127.0.0.1:3000/hafsatar iq071/Assignment01.git'
hafsatar iq071@ubuntu:~/Assignment01$
```

task1 gitea push.png

Step 1.3: Verify the Gitea Repository



task1 gitea repository.png

Task 2: Push the Same Repository to GitHub

Step 2.1: Create the GitHub Repository and Verify Both Remotes

1.

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner *

 hafsatriq071

Repository name *

Assignment071

✔ Assignment071 is available.

Great repository names are short and memorable. How about [bug-free-pancake?](#)

Description

0 / 350 characters

2 Configuration

Choose visibility *

Choose who can see and commit to this repository

 Public

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

Off ☐

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore

Add license

Licenses explain how others can use your code. [About licenses](#)

No license

2.

 GitHub Apps

 OAuth Apps

 Personal access tokens

Fine-grained tokens

Tokens (classic)

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Note

Ubuntu assignment

What's this token for?

Expiration

 30 days (Nov 03, 2026)

The token will expire on the selected date

Select scopes

Scopes define the access for personal tokens. [Read more about OAuth scopes](#).

4.

```
hafsatarig071@ubuntu:~/Assignment01$ git remote add github https://github.com/hafsatarig071/Assignment01.git
```

5.

```
hafsatarig071@ubuntu:~/Assignment01$ git remote -v
gitea  http://127.0.0.1:3000/hafsatarig071/Assignment01.git (fetch)
gitea  http://127.0.0.1:3000/hafsatarig071/Assignment01.git (push)
github https://github.com/hafsatarig071/Assignment01.git (fetch)
github https://github.com/hafsatarig071/Assignment01.git (push)
hafsatarig071@ubuntu:~/Assignment01$
```

task2_remotes.png

Step 2.2: Push to GitHub

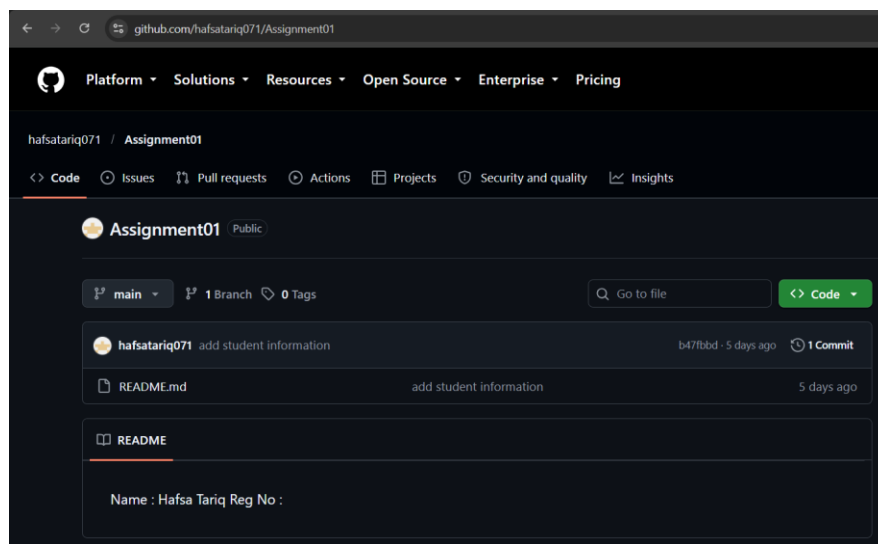
```
hafsatarig071@ubuntu:~/Assignment01$ git remote set-url github https://hafsatarig071:ghp_2yUN6e74rmQJdmJHmX3uAT9xpg7wJh3qJ72e@github.com/hafsatarig071/Assignment01.git
hafsatarig071@ubuntu:~/Assignment01$ git push -u github main
```

```
fatal: repository 'https://github.com/hafsatarig071/Assignment01.git/' not found
hafsatarig071@ubuntu:~/Assignment01$ git push -u github main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 262 bytes | 26.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/hafsatarig071/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
hafsatarig071@ubuntu:~/Assignment01$
```

t input to this VM, click inside or press Ctrl+G.

task2_github_push.png

Step 2.3: Verify the GitHub Repository



task2_github_repository.png

Task 3: Track Three Large Files with Git LFS

Step 3.1: Configure Git LFS

```
git-lfs --help (GitHub; linux amd64; go 1.25.0)
hafsatarig071@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
hafsatarig071@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
hafsatarig071@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
hafsatarig071@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
hafsatarig071@ubuntu:~/Assignment01$
```

task3 lfs setup.png

Step 3.2: Create and Verify Three LFS Files

```
hafsatarig071@ubuntu:~/Assignment01$ truncate -s 101M large-file-1.bin
hafsatarig071@ubuntu:~/Assignment01$ truncate -s 102M large-file-2.bin
hafsatarig071@ubuntu:~/Assignment01$ truncate -s 103M large-file-3.bin
hafsatarig071@ubuntu:~/Assignment01$ git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
hafsatarig071@ubuntu:~/Assignment01$ git lfs ls-files --size
f1753a094b * large-file-1.bin (106 MB)
2338a970c0 * large-file-2.bin (107 MB)
cb91d5908e * large-file-3.bin (108 MB)
hafsatarig071@ubuntu:~/Assignment01$
```

task3 lfs files.png

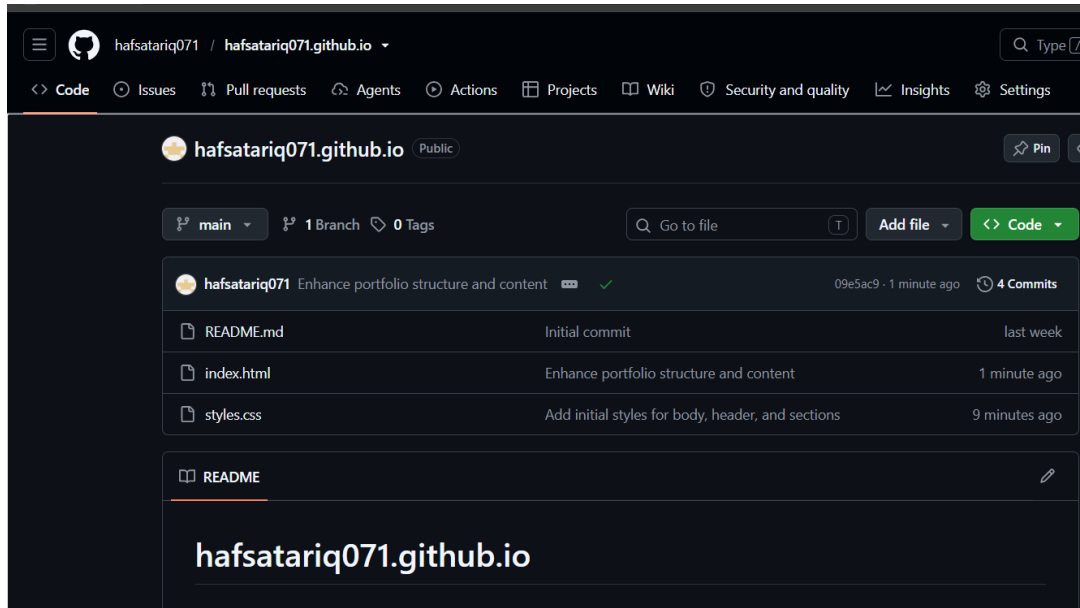
Step 3.3: Commit and Push the LFS Files

```
hafsatarig071@ubuntu:~/Assignment01$ git commit -m "add three files using Git LFS"
[main 1b88823] add three files using Git LFS
 4 files changed, 10 insertions(+)
 create mode 100644 .gitattributes
 create mode 100644 large-file-1.bin
 create mode 100644 large-file-2.bin
 create mode 100644 large-file-3.bin
hafsatarig071@ubuntu:~/Assignment01$ git push github main
Uploading LFS objects: 100% (3/3), 321 MB | 2.5 MB/s, done.
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 794 bytes | 88.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/hafsatarig071/Assignment01.git
 b47fbbd..1b88823 main -> main
hafsatarig071@ubuntu:~/Assignment01$
```

task3 lfs push.png

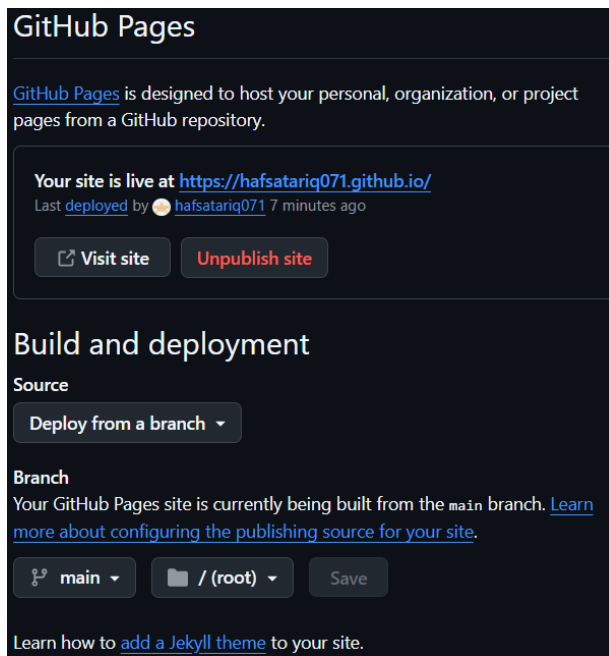
Task 4: Create a Portfolio or CV with GitHub Pages

Step 4.1: Create and Push the Portfolio Repository



task4_pages_repository.png

Step 4.2: Publish with GitHub Pages



task4_pages_deployment.png

Step 4.3: Verify the Live Portfolio



task4 portfolio live.png